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# Winter cover crops in Dutch maize fields: Variability in quality and its drivers assessed from multi-temporal Sentinel-2 imagery

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ABSTRACT

Planting a cover crop between the main cropping seasons is an agricultural management measure with multiple potential benefits for sustainable food production. In the maize production system of the Netherlands, an effective establishment of a winter cover crop is important for reducing nitrogen leaching to groundwater. Cover crop establishment after maize cultivation is obliged by law for sandy soils and consequently implemented on nearly all maize fields, but the winter-time vegetative ground cover varies significantly between fields. The objectives of this study are to assess the variability in winter vegetative cover and evaluate to what extent this variability can be explained by the timing of cover crop establishment and weather conditions in two growing seasons (2017–2018). We used Sentinel-2 satellite imagery to construct NDVI time series for fields known to be cultivated with maize within the province of Overijssel. We fitted piecewise logistic functions to the time series in order to estimate cover crop sowing date and retrieve the fitted NDVI value for 1 December ($NDVI_{Dec}$). We used $NDVI_{Dec}$ to represent the quality of cover crop establishment at the start of the winter season. The Sentinel-2 estimated sowing dates compared reasonably with ground reference data for eight fields (RMSE = 6.6 days). The two analysed years differed considerably, with 2018 being much drier and warmer during summer. This drought resulted in an earlier estimated cover crop sowing date (on average 19 days) and an $NDVI_{Dec}$ value that was 0.2 higher than in 2017. Combining both years and all fields, we found that Sentinel-2 retrieved sowing dates could explain 55% of the $NDVI_{Dec}$ variability. This corresponded to a positive relationship ($R^2$ = 0.50) between $NDVI_{Dec}$ and the cumulative growing degree days (GDD) between sowing date and 1 December until reaching 400 GDD. Based on cumulative GDD derived from two weather stations within Overijssel, we found that on average for the past three decades a sowing date of 19 September ( ± 7 days) allowed to attain these 400 GDD; this provides support for the current legislation that states that from 2019 onwards a cover crop should be sown before 1 October. To meet this deadline, while simultaneously ascertaining a harvest-ready main crop, in practice implies that undersowing of the cover crop during spring will gain importance. Our results show that Sentinel-2 NDVI time series can assess the effectiveness and timing of cover crop growth for small agricultural fields, and as such has potential to inform regulatory frameworks as well as farmers with actionable information that may help to reduce nitrogen leaching.

## 1. Introduction

Cover crops, which are grown after a main crop to provide soil cover during winter can reduce nitrogen leaching to groundwater by accumulating residual soil nitrogen into their biomass (Aronsson et al., 2016; Plaza-Bonilla et al., 2015; Singer et al., 2011). Cover crops also improve soil health by increasing soil organic matter (Ding et al., 2006), preventing erosion by water or wind (Estler, 1991), suppressing weeds (Wells et al., 2014) and pests (Liburd et al., 2008), conserving moisture (Munawar et al., 1990), and increasing soil nutrient availability for the following summer crops due to nitrogen fixation and green manuring (Rosecrance et al., 2000).

For more than twenty years, Dutch farmers have had a legal obligation to grow cover crops after maize cultivation on sandy and loess soils (Brussaard, 1992). The main purpose of this obligation is to reduce nitrogen leaching into the groundwater. Leaching poses great risks to the drinking water supply (Schröder et al., 2004; Wick et al., 2012) and deteriorates aquatic ecosystems, resulting in eutrophication, algal blooms, and fish poisoning (Nieder and Benbi, 2008). The European Union's Nitrates Directive sets the acceptable threshold of nitrate concentration in groundwater at 50 mg/L (Council Directive 91/676/EEC, 1991). While Dutch farmers have taken efforts to control nitrate leaching, resulting in an average decrease of nitrate concentration in

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groundwater during the past twenty years, the Dutch National Institute for Public Health and the Environment (RIVM) reported for 2012–2015 that in multiple locations nitrate concentrations in the Netherlands continue to be above this threshold (Fraters et al., 2017). Therefore, the Dutch government has intensified compliance checks and is adapting the regulatory framework to ascertain effective cover crop establishment. Non-compliance with these rules can result in administrative fines and reduced subsidies.

Although as a result of the intensified compliance checks cover crops are established on nearly all maize fields, the vegetative ground cover in winter varies significantly between fields, likely resulting from a range of factors such as sowing date, weather conditions, species choice, seeding rate, planting method, and the remaining soil nitrogen after the maize crop is harvested (Duiker, 2014; Hively et al., 2015; Lawson et al., 2015). While both root and aboveground biomass of cover crops are responsible for the uptake of excess nitrogen from the soil, for winter rye the largest fraction (85%) of total nitrogen uptake is partitioned to the aboveground biomass (Brennan and Boyd, 2012; Patel et al., 2015), indicating that low aboveground biomass in the winter season greatly increases the risk of nitrogen leaching.

The earlier the cover crop is sown, the greater the chances that it can provide abundant vegetative ground cover in winter (Lawson et al., 2015; Mirsky et al., 2011), even at lower seeding rates (Lloveras et al., 2004). For example in Pennsylvania, biomass of rye increased by about 50% when planted in early September instead of mid-October (Mirsky et al., 2011). Similarly in Washington State, Lawson et al. (2015) found that a 2.5 week delay in planting rye reduced average winter ground cover by 65%, biomass by 50% and crop nitrogen accumulation by 40%.

To improve cover crop establishment, from 2019 onwards the Netherlands implemented a new regulation for maize production systems that requires cover crops on sandy and loess soils to be sown no later than 1 October. This is a challenge for silage maize growers given that the desired ripeness, which is determined by a dry matter ratio of 34%–40% (van Schooten et al., 2019), is on average not yet reached by that date, potentially resulting in sub-optimal maize harvests. Despite the evidence of the importance of early sowing, location-specific information on optimal sowing dates is needed, in order to evaluate 1) whether the current date of 1 October is sensible, 2) what management options farmers have to attain a good winter vegetative cover.

For temperate and continental climates with cold winters, temperature is the most important weather parameter affecting cover crop establishment. Due to their cold tolerance (Bredow et al., 2017; Thomas and James, 1993), rye (*Secale cereale*) and Italian ryegrass (*Lolium multiflorum*, also referred to as *Festuca perennis*) are the two most common cover crops used after maize is harvested in the Netherlands. While a warm autumn is crucial for the germination and growth of these cover crops (Wilson et al., 2013), winter survival is not hampered by cold winters (Feyereisen et al., 2006), and temperature has been found to be more important for seedling emergence than soil moisture (Blackshaw, 1991; Cutforth et al., 1985). Therefore growing degree days (GDD), i.e., the accumulation of daily heat units from sowing date, is a useful measure to simulate (cover) crop biomass development (Feyereisen et al., 2006; Mirsky et al., 2009). As a consequence, GDD may be used to assess expected cover crop biomass differences due to different sowing dates.

Although GDD can provide a proper modelling framework for assessing the effect of sowing dates on cover crop performance, we also require observational data. While field trials (Lawson et al., 2015; Mirsky et al., 2009; Wilson et al., 2013) are useful for this purpose, their site-specific nature may not allow for generalisation across a wider landscape with variability in weather conditions, farm management, and soil properties. An alternative approach is the use of satellite imagery. Besides assessing the winter status of the cover crop (Hively et al., 2015; Seifert et al., 2018), new generation satellite systems with short observation intervals, like Sentinel-2, can estimate seasonality parameters at sufficient spatial detail to monitor individual fields (e.g., Pan et al., 2015; Stendardi et al., 2019; Vrieling et al., 2018; Liu et al., 2018).

Focusing on the application of these new generation satellite technologies, the objectives of this study are to assess the variability in winter vegetative cover and to evaluate to what extent this variability can be explained by the timing of cover crop establishment and weather conditions in two growing seasons (2017–2018). We took advantage of the capability of Sentinel-2 to collect observational data on crop cover sowing time and vegetative ground cover at the start of the winter season across a large area. The Sentinel-2 derived retrievals were subsequently used to evaluate the link between the timing of cover crop establishment, the winter vegetative ground cover, and temperature, focussing on the province of Overijssel across two consecutive growing seasons. Based on this analysis, we then inferred optimal cover crop sowing times with the aim of assisting in the definition of options for improved establishment of cover crops that may help to reduce nitrogen leaching.

## 2. Study area and data

### 2.1. Study area

This study focused on maize fields within the province of Overijssel (3326 km$^2$) located in the eastern part of the Netherlands (Fig. 1). Its land use consists of 61% agricultural land (dominated by dairy farming

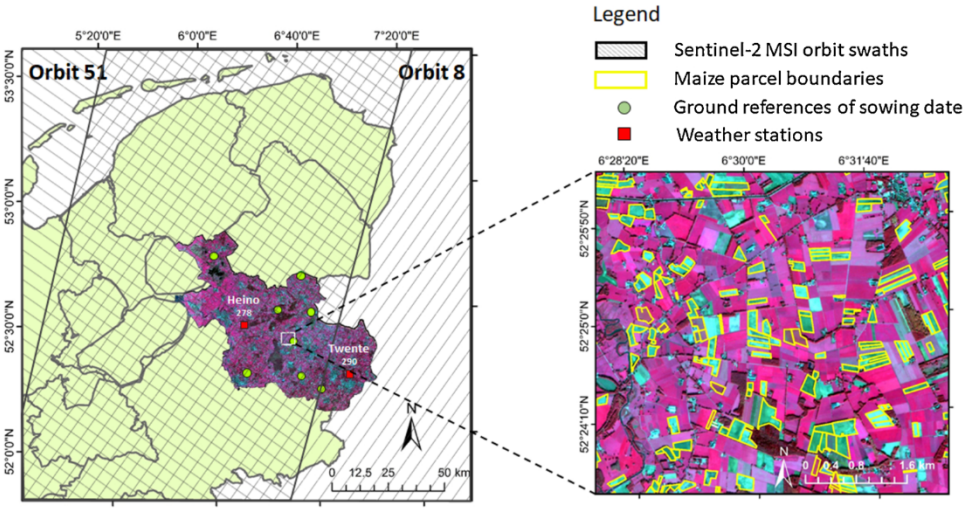



**Fig. 1.** Study area in the province of Overijssel covering two Sentinel-2 MSI orbit swaths, with locations of weather stations and fields for which crop cover sowing dates were available. The righthand figure shows a detail of the false-colour (NIR – Red – Green) Sentinel-2 satellite image collected on 15 October 2017. (For interpretation of the references to colour in this figure citation, the reader is referred to the web version of this article.)

with 80% grassland and 20% maize), 27% natural areas including forest, heather, and peatlands, and 12% urban areas, according to the national land use dataset for 2015 obtained from Statistics Netherlands [https://data.overheid.nl/dataset/58880-bestand-bodemgebruik-2015]. Silage maize, used for cattle feeding, makes up 98% of the total maize area in the province (Dutch Basic Registration Parcels [Dutch Basic Registration Parcels BRP, 2017]). The average size of the maize fields is 2.4 ha ( ± 1.8 ha). Approximately 90% of the maize fields are on sandy soils (i.e., with a cover crop obligation) and their groundwater table is typically 2 − 4 m below surface. The remaining fields are on peat (3%) and clay (7%) soils (Dutch Soil Information System, 2019). The study region has a temperate oceanic climate with a regional average annual temperature of 9.3 °C. The average annual precipitation is 785 mm and falls throughout the year. The cover crops used are dominated by rye and Italian ryegrass, although some farmers started to apply cover crop mixtures because these proved to be more productive than their component monocultures (Barel et al., 2018). The sowing of the cover crop normally takes places as soon as possible after the maize harvest, i.e., generally within one week.

### 2.2. Sentinel-2 imagery

The multispectral instrument (MSI) is flown on two identical satellites: Sentinel-2A that was launched on 23 June 2015 and Sentinel-2B that was launched on 7 March 2017. We acquired all (partially) cloud-free Sentinel-2A and -2B images for Overijssel between May and December for the years 2017 and 2018. This resulted, per tile, in 37 images for 2017 and 50 for 2018. We note that the first available Sentinel-2B image for Overijssel was acquired on 6 August 2017 which explains the reduced image availability in 2017. The combination of both satellites results in a five-day revisit time, but towards the poles there is an increasing area of swath overlap, resulting in two observations per five days. For the larger part of Overijssel this was the case, resulting at a larger number of (cloud-free) observations in that region as compared to south-east Overijssel (Fig. 1). Atmospheric correction and cloud masking were performed by the French inter-agency Theia Land Data Centre [www.theia-land.fr] with the Multi-sensor Atmospheric Correction and Cloud Screening ([MACCS], Hagolle et al., 2015) spectro-temporal processor. MACCS uses the Simplified Model for Atmospheric Correction (SMAC, Rahman and Dedieu, 1994) algorithm for atmosphere absorption correction, and applies multi-temporal criteria for improved cloud detection and aerosol retrieval. We decided to use the MACCS dataset rather than the surface reflectance products by ESA (generated with Sen2Cor) because of the reported improved accuracy for cloud and shadow detection (Baetens et al., 2019) with similar NDVI results (Sola et al., 2018). For each image, pixels identified by MACCS as saturated, snow, shadow, or cloud were discarded in our further analyses. For each retained observation, the normalized difference vegetation index (NDVI: Rouse et al., 1974) was calculated using the red spectral band (band 4) and the near infrared band (band 8), as NDVI was reported to outperform many other indices when assessing vegetative ground cover across cover crop fields (Prabhakara et al., 2015). Both spectral bands have a 10 m spatial resolution, resulting in a 10-m NDVI image time series for the province.

### 2.3. Ancillary data

Maize field boundaries within Overijssel were obtained from the Dutch BRP dataset provided by the Ministry of Economic Affairs and Climate Policy [http://geodata.nationaalgeoregister.nl/brpgewaspercelen/atom/brpgewaspercelen.xml]. The BRP collects the location and boundaries of all agricultural fields, and the farmer-reported crops that were cultivated on those fields each year. Part of the reported crops were subsequently verified by the Ministry using high resolution satellite imagery and field inspection. Based on this and our own checks in the field, we can safely assume a high-quality crop labelling from this dataset. The BRP dataset is annually updated, and contains 13,921 maize fields for 2017 and 14,816 for 2018 within Overijssel. To ascertain that the NDVI captures information from the field, rather than incorporating signals of the surroundings (e.g., tree shadows, drainage ditches), we decreased the size of each field by applying a 5 m internal buffer. The average number of retained pixels after buffering for the maize fields is 223 pixels ( ± 178 pixels).

Daily gridded minimum and maximum temperature for the years 2017 and 2018 were obtained from the Royal Netherlands Meteorological Institute (KNMI). These data have a spatial resolution of 20 km and are produced based on 35 automatic weather stations across the Netherlands. Two of these stations, i.e., 'Twente' and 'Heino', are within the province of Overijssel (Fig. 1), and we also used their station data directly for 1989–2018.

We obtained cover crop sowing date reference data for eight fields in 2017 from a questionnaire survey held among farmers. Although questionnaires gathering information about farm management practices were filled for more fields during the survey, only eight reported the exact date instead of the approximate time period (1–2 weeks of the month) of cover crop sowing (Fig. 1).

## 3. Methods

### 3.1. Phenology retrieval from Sentinel-2 NDVI time series

Field-level NDVI time series were generated by spatially averaging Sentinel-2 NDVI values contained within each 5 m-buffered field boundary. For each maize field, a piecewise curve fitting approach was used to smooth the NDVI series. Curve fitting is widely used to model vegetation phenology due to its effectiveness in reducing the impact of remaining noise in the data (Wang et al., 2017). The 'piecewise' element consisted of fitting separate double logistic functions to two (overlapping) NDVI time series, i.e., 1) the increase and subsequent decrease of NDVI corresponding to the maize growth and harvest, and 2) the decrease and increase of NDVI corresponding to the maize harvest and cover crop growth (Fig. 2). Two local functions were fitted to model the important elements of the growth cycles, because the model functions perform well in broad intervals around maxima and minima, while the fits are less good at the limbs (Eklundh and Jönsson, 2015). Although the maize growth trajectory may not strictly be required for this study that focuses on the cover crop, we preferred to assess the full maize/cover crop cycle because the timing of cover crop sowing and establishment strongly depends on the preceding maize growing season. Model fitting was only attempted if the field-level NDVI time series had at least two valid NDVI values between 1 September and 1 November (corresponding to the common period of cover crop sowing). The NDVI time series generally showed two peaks, i.e., during maximum photosynthesis of maize and of the cover crop before winter dormancy.

Each double logistic function is of the form (Elmore et al., 2012):

$$f(t) = m_1 + (m_2 - m_7 t)\left(\frac{1}{1 + e^{(m_3 - m_4 t)}} - \frac{1}{1 + e^{(m_5 - m_6 t)}}\right) \quad (1)$$

where $t$ is time (days). The model parameters are:

- $m_1$: the overall minimum (base) NDVI value,
- $m_2$: amplitude from the base to the maximum,
- $m_3$ [$m_5$]: the inflection point for onset [offset] of greenness,
- $m_4$ [$m_6$]: the slope of the first [second] half of one local time series
- $m_7$: an additional linear slope that can account for effects like the greenness reduction of maize during summer (i.e., before harvest).

To find optimal parameters for each NDVI field-level time series, we fitted the functions with the least-squares inversion method (Menke, 1989) and started the optimisation from 3000 sets of initial parameters in R 3.5.0 (R Core Team, 2018), using the *greenbrown* package (*v2.2*; Forkel and Wutzler, 2015). Finally, both local functions were merged to obtain a single global function, within the overlapping period from the maximum ($t_L$) of the left local function ($f_L(t)$) and the minimum ($t_R$) of

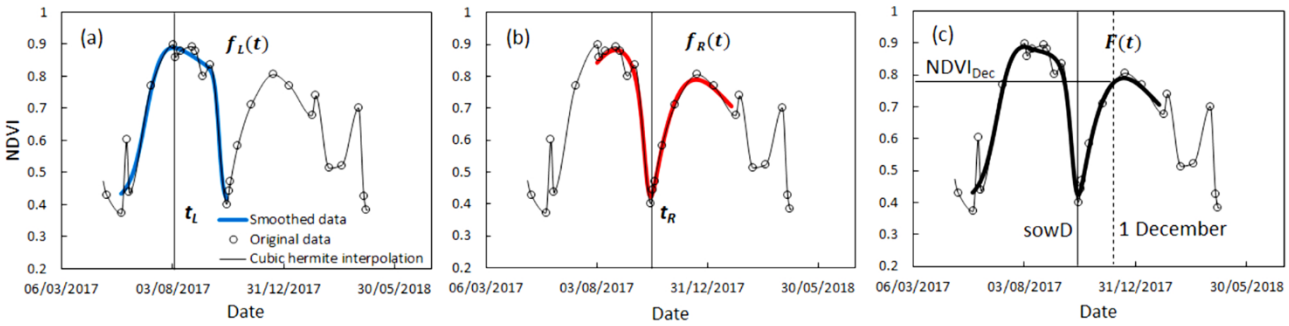



**Fig. 2.** Two local double-logistic functions were fitted to describe (a) maize growth and decline, and (b) maize decline and cover crop growth. Both local functions were merged to obtain a single global function from which two parameters (sowD and $NDVI_{Dec}$) were extracted (c).

the right local function ($f_R(t)$):

$$F(t) = \alpha(t) f_L(t) + (1 - \alpha(t)) f_R(t) \quad t_L < t < t_R \tag{2}$$

where α(t) linearly drops from 1 at $t_L$ to 0 at $t_R$ (Eklundh and Jönsson, 2015). This is illustrated in Fig. 2.

From the global function, we extracted two parameters. The first is the estimated cover crop sowing date (sowD), which we obtain as the time of fitted minimum NDVI. The second is $NDVI_{Dec}$, the fitted NDVI value for 1 December (Fig. 2c). We selected $NDVI_{Dec}$ to represent the quality of the cover crop establishment at the start of the winter season. This selection was justified by the fact that 1 December marks the start of the meteorological winter. In the three subsequent months, average monthly temperatures are below 4 °C and thus less conducive to cover crop growth. December and January are also on average among the wettest months of the year, hence potentially have a high risk of nitrogen leaching. In addition, the NDVI series are observed to be noisier due to persistent clouds and potential snow cover after 1 December.

### 3.2. Explanatory factors for winter vegetative cover variability

Although we acknowledge the possible importance of other factors for winter vegetative cover variability (as expressed by $NDVI_{Dec}$), in this study we focussed only on time-variant factors. These include cover crop sowing date and temperature. We confined our analysis to fields for which the temporal NDVI information was sufficient to effectively describe the maize and cover crop growth cycle, particularly during the transition from maize harvest to cover crop growth. This was achieved by examining two conditions: (1) the maximum gap of observations (maxGap) in a time window ranging from 30 days before to 15 days after sowD should not be larger than 20 days, (2) the coefficient of determination ($R^2$) between the observed and fitted NDVI values for each time series should be at least 0.95.

To assess to what extent variability in winter vegetative cover can be explained by the timing of cover crop establishment, we empirically modelled the relationship between sowD and $NDVI_{Dec}$ for all the retained maize fields for 2017 and 2018 separately, and for both years combined. A logistic regression model was selected based on visual inspection of the scatterplot between the two variables. The predictive ability of the empirical models was evaluated using the root-mean-square error (RMSE) and $R^2$.

Given the importance of temperature for cover crop development during autumn, we analysed the seasonal mean temperature between sowD and 1 December in relation to $NDVI_{Dec}$. To achieve this, we first divided sowD into six groups of 10-day intervals, each group having approximately the same sowD and thus the same growth period from sowD to 1 December. For each group, we then calculated the spatio-temporal mean of the season temperature using the gridded temperature data for each field between average sowD and 1 December. Each of the six groups was then divided into two sub-groups, i.e., those with a mean temperature above, and those with a mean temperature below the mean, to assess if the sub-groups showed a significant difference in $NDVI_{Dec}$.

### 3.3. Assessing optimal cover crop sowing time from multi-year temperature data

To understand the combined influence of sowD and temperature on $NDVI_{Dec}$, we calculated the GDD, a measure of heat accumulation, between sowD and 1 December for both years and all maize fields. The field-level GDD was calculated as the difference between the average of daily maximum ($T_{max}$) and minimum ($T_{min}$) temperature extracted from the gridded daily temperature dataset, and a base temperature ($T_{base}$) (Miller et al., 1990):

$$GDD = \max\left(\left(\frac{T_{max} + T_{min}}{2} - T_{base}\right), 0\right) \tag{3}$$

Although $T_{base}$ varies based on species, plant variety, and the purpose of the analysis itself (Wypych et al., 2017), for winter small grains, GDD is often calculated with $T_{base}$ of either 0 or 4 °C. While 0 °C is commonly used for cool-season crops (e.g., Poffenbarger et al., 2015), for rye 4 °C was found to provide a better link between GDD and both winter ground cover and aboveground biomass (Prabhakara et al., 2015; Lawson et al., 2015). Therefore, for the cover crops in Overijssel (i.e., principally rye and Italian ryegrass) we selected 4 °C as $T_{base}$. Cumulative GDD was calculated by accumulating daily GDD units from sowD through 1 December.

Combining data from both seasons and all retained maize fields, we analysed the relationship between cumulative GDD and $NDVI_{Dec}$. This allowed identification of a threshold value for cumulative GDD, above which $NDVI_{Dec}$ remained at a similar high level, and below which $NDVI_{Dec}$ showed a strong decrease. Data from the two operational weather stations in Overijssel were then used to identify, for each year in the 1989–2018 period, what the sowing date for the cover crop should have been in order to attain the cumulative GDD threshold level by 1 December.

## 4. Results

### 4.1. Phenology retrievals from Sentinel-2 NDVI time series

Fig. 3 shows the Sentinel 2-derived NDVI time series for the eight reference fields, together with the fitted functions and the two extracted parameters (sowD and $NDVI_{Dec}$) for 2017 and 2018. No NDVI time series are shown for 2018 for Fig. 3g because no maize was cultivated in that year (the field changed to grassland).

For all the fields in Fig. 3, there was a reasonable spread of Sentinel-2 observations between the maize planting in May until the peak of cover crop growth in December, with an average observation interval of 13.9 days and 8.6 days in season 2017 and 2018, respectively. This interval is smaller for 2018, because despite that Sentinel-2B was launched in March 2017, the earliest available Sentinel-2B image for Overijssel is acquired in August 2017. The piecewise smoothing method allowed to effectively model the NDVI temporal behaviour for all the

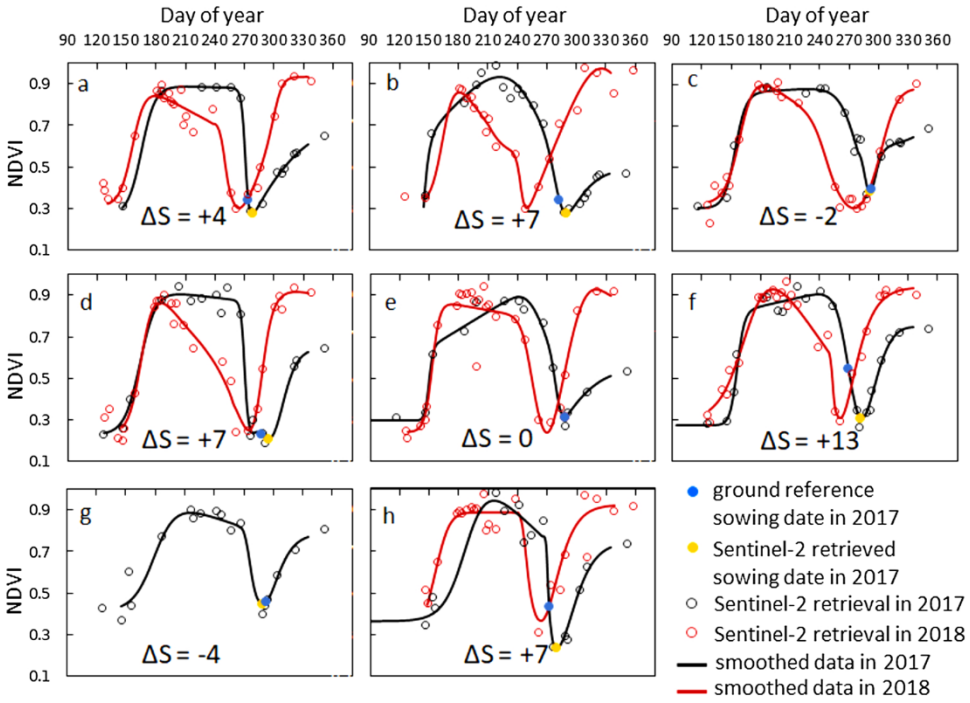



**Fig. 3.** Cover crop sowing date retrievals for eight maize-cover crop rotated fields with ground reference data. ΔS is the absolute difference between ground reference and Sentinel-2 retrieved sowing date. '+' means that retrieved date is later than the farmer-reported date.

eight fields, with a very high average coefficient of determination ($R^2$ = 0.97) between original and fitted NDVI values.

The sowD retrieved from Sentinel-2 is on average later than the farmer-reported ground reference in 2017, with a mean signed difference (MSD) of 4 days (Fig. 3). This average delay is mostly due to the fact that cover crop emergence also has a delay with respect to sowing. For field 'f', the absolute error (ΔS = + 13) of the sowD estimate was substantially larger than for other fields. This may be explained by the limited temporal NDVI information during the transition from maize harvest to cover crop growth due to the large maximum gap of observations within the transition period (i.e., a big gap after the last high value, maxGap = 25 days). For example, fields 'c' and 'e' had a maximum gap of 15 and 12 days, respectively, and have sowD estimation errors of two days or less. Despite small discrepancies, the piecewise-based smoothing method detected cover crop sowD with low root-mean-squared error (RMSE = 6.6 days), suggesting that the Sentinel-2 retrieved sowD is an effective proxy of the real sowing date for the agricultural fields in Overijssel. Fig. 3 also shows that the hot, dry summer of 2018 resulted in a shorter period of maize growth and an earlier cover crop sowD (for Fig. 3a – f on average 19 days). The 2018 $NDVI_{Dec}$ values for the six reference fields were on average 0.32 higher than in 2017.

### 4.2. Variability in winter vegetative cover and cover crop sowing date

Fig. 4 shows the field-level retrievals of cover crop phenology from Sentinel-2 for all maize fields within Overijssel in the growing season 2017 – 2018. The retrievals succeeded for 83% and 94% of the fields in 2017 and 2018, respectively. For the remaining fields a piecewise model could not be fitted because the criterion for performing a model fit was not met (3.2% and < 0.1% of the fields did not have the minimum of two NDVI observations in the September – November period in 2017 and 2018, respectively) or the model optimisation failed to converge (13.6% of the fields in 2017 and 5.8% in 2018). Regarding the criterion for model fit, we note that on average for all the maize fields 5.5 and 6.5 valid (cloud-free) NDVI observations were available for September-November in 2017 and 2018, respectively. The convergence failure often happened due to large temporal gaps between consecutive cloud-free NDVI values. For 2017, fewer successful retrievals were obtained, which can be attributed a) to the smaller number of Sentinel-2 observations due to the absence of Sentinel-2B images for a large part of the season, and b) to more cloud cover during the maize-cover crop transition period. The spatial distribution of fields where phenology retrievals failed can be found in Fig. 5. For areas not imaged by overlapping orbits (eastern Overijssel), phenology retrieval had a lower success rate as compared to the whole province, i.e., 37% for the 2017 fields, and 83% for 2018.

For the north of the province, crop cover sowing dates are on average later and consequently has lower $NDVI_{Dec}$ than in the south for both years (Fig. 4). The later sowing dates relate to later maize harvests, which in turn are due to colder temperatures during maize growth and consequently a longer period is required to achieve maize ripeness (Fig. S2). Comparing 2017 and 2018 for all maize fields, the dry and hot summer of 2018 resulted in earlier cover crop sowing dates (on average 19 days) and an $NDVI_{Dec}$ that was on average 0.21 larger than in 2017. The insets of Fig. 4 illustrate that sowing dates covered a wider temporal range in 2018 than in 2017, because farmers made different decisions when faced with significant drought impacts on their fields (e.g., early harvesting of a non-mature but drought-affected maize crop). In addition, the larger North – South temperature difference during the maize growth (May – September) in 2018 may also cause a wider temporal range of maize harvest and subsequent cover crop sowing dates (Fig. S2).

### 4.3. Effect of sowing date and temperature on winter ground cover

Fig. 6 displays box-and-whisker plots of $NDVI_{Dec}$ against sowD for two growing seasons (2017 – 18), together with the result of the logistic regression. Of all the field-level phenology retrievals, 5959 and 6220 fields with detailed NDVI information around sowing date and effective model fits were retained for plotting for 2017 and 2018, respectively. 31.6% and 25.0% of the fields were discarded because maxGap exceeded 20 days, and 16.9% and 30.4% due to sub-optimal $R^2$ values (< 0.95) between observed and fitted NDVI in 2017 and 2018,

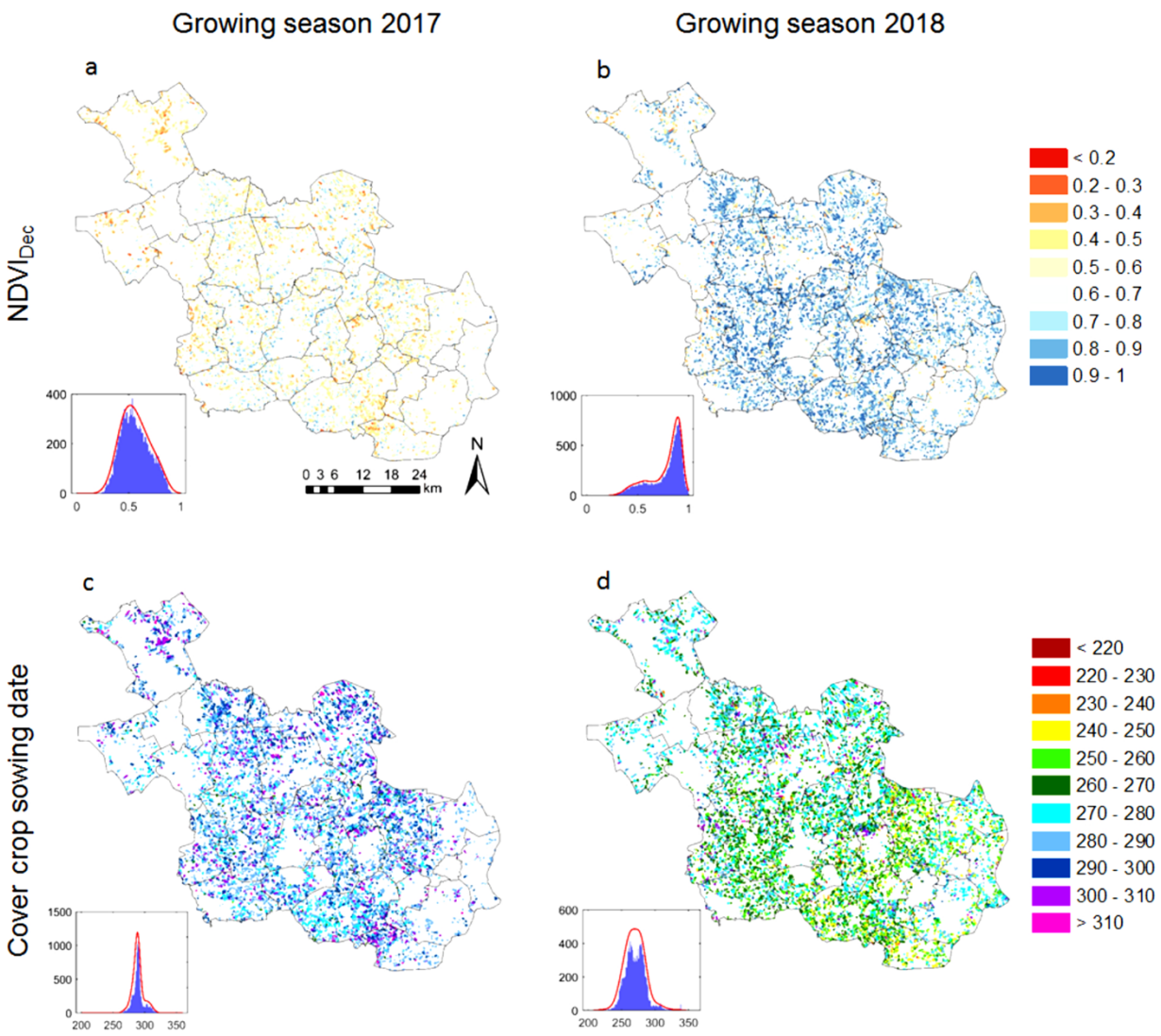



**Fig. 4.** Phenology results for maize fields in Overijssel: spatial variations of $NDVI_{Dec}$ in 2017 (a) and 2018 (b); cover crop sowing date (day of year) in 2017 (c) and 2018 (d), along with insets of their frequency distributions.

respectively. Additionally, in Fig. 6c approximately 3% of the observations were considered outliers because they exceed 1.5 times the interquartile range. Those observations predominately correspond to fields with early sowing dates (255–270 DOY), but still with low $NDVI_{Dec}$. This group of outliers can potentially be explained by farm management given that no specific spatial clustering of these outlier fields was found, such as the use of very low seeding rates, and possibly low fertilisation rates resulting in low remaining soil nitrogen levels

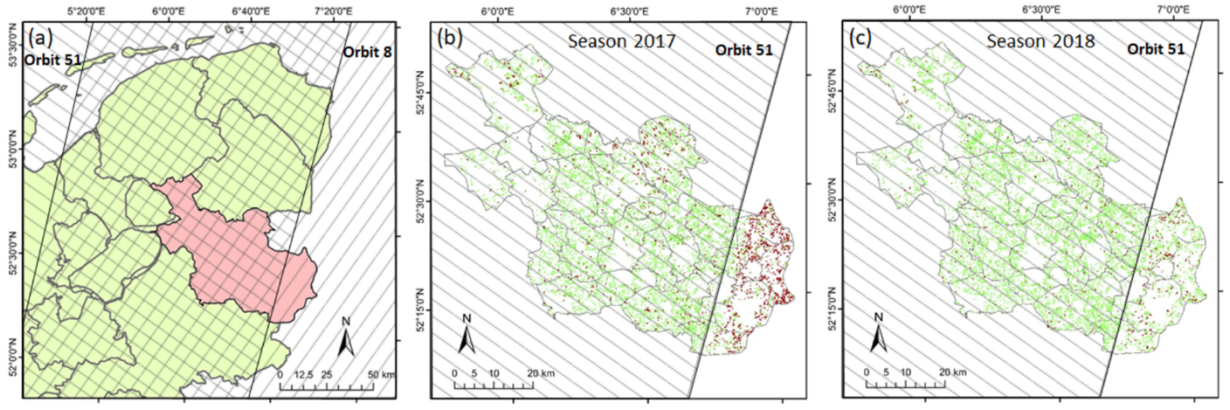



**Fig. 5.** Study area (a) in the province of Overijssel covering two Sentinel-2 MSI orbit swaths. (b) and (c) illustrate maize fields with successful (green) and failed (red) phenology retrievals in growing season 2017 and 2018, respectively. (For interpretation of the references to colour in this figure citation, the reader is referred to the web version of this article.)

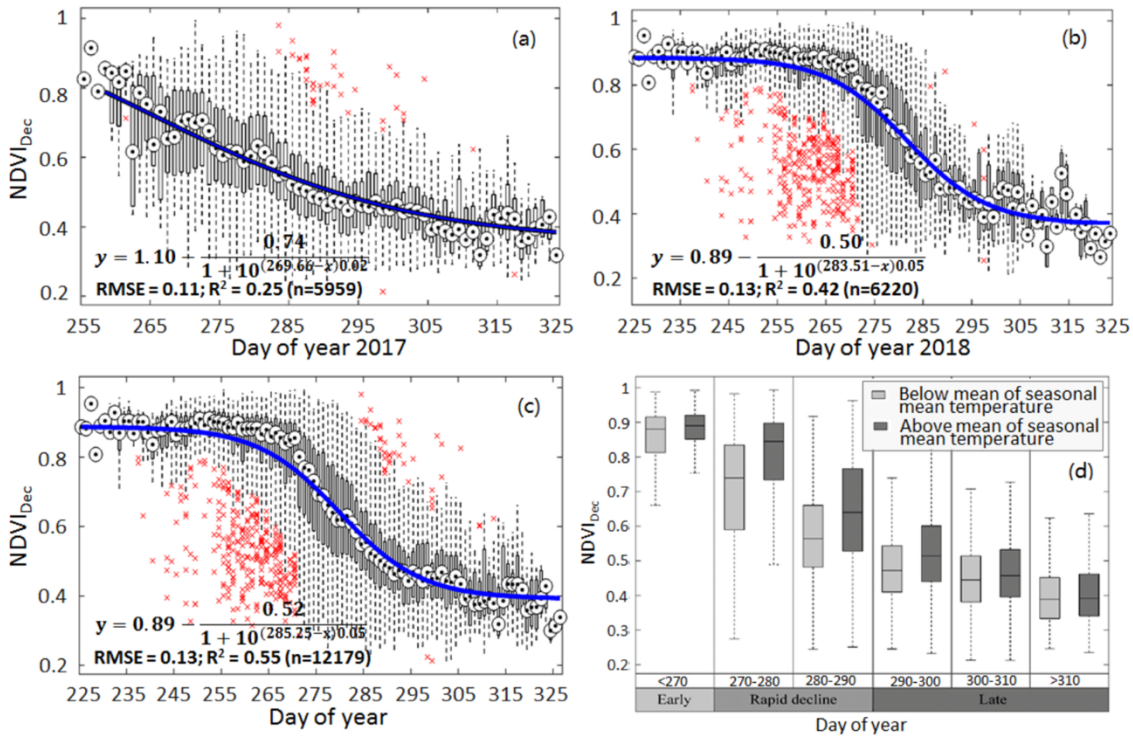



**Fig. 6.** Box-and-whisker plot between $NDVI_{Dec}$ and cover crop sowing date for growing season 2017(a), 2018 (b) and both seasons (c), together with their logistic regression results. Each box corresponds to a single day, whereby the central mark (dot-in-circle) indicates the median, and the bottom and top edges of the box indicate the 25th (Q3) and 75th (Q1) percentiles, respectively. Observations were considered outliers if they are more than 1.5 × (Q3 − Q1) from the edge of the box. Outliers are plotted individually using a red '×'. (d) shows box-and-whisker plot of $NDVI_{Dec}$ (categorised by the seasonal mean temperature) at different growth periods (i.e., 10-day sowD intervals). For each growth period, two $NDVI_{Dec}$ sub-datasets are created by taking the mean of the seasonal mean temperature of one maize field as the threshold. (For interpretation of the references to colour in this figure citation, the reader is referred to the web version of this article.)

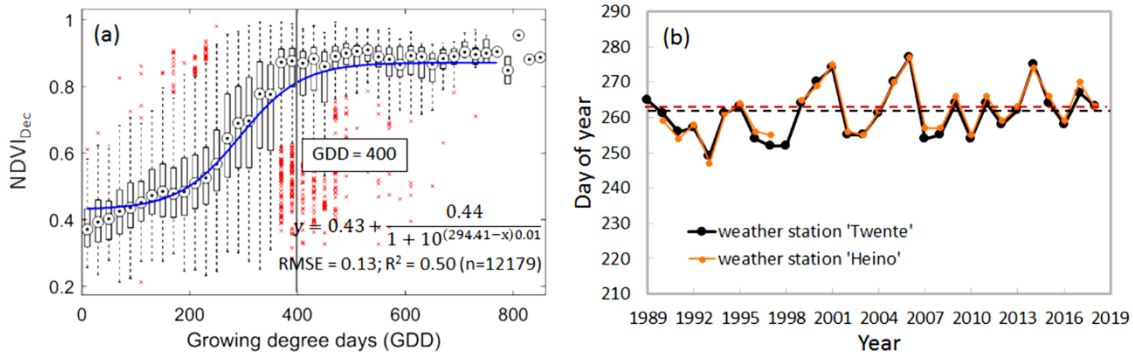



**Fig. 7.** (a) Box-and-whisker plot between $NDVI_{Dec}$ and cumulative GDD, with the logistic regression results. $NDVI_{Dec}$ increases rapidly until cumulative GDD reaches 400. (b) shows per year on which date a cover crop should have been sown to reach a cumulative GDD of 400, calculated using temperature data recorded at weather stations 'Twente' and 'Heino' for the past three decades. Dashed lines represent the average of optimal cover crop sowing dates.

after maize harvest. Besides, despite the assumed high accuracy, small crop labelling errors in the BRP could be an additional cause for some of the outliers.

For both years, the logistic regression curves accurately described the relationship between median $NDVI_{Dec}$ and sowD (Fig. 6). Due to the wider range of sowD in 2018, the logistic model showed a full inverted S-shaped curve (Fig. 6b), representing a nonlinear relationship between $NDVI_{Dec}$ and sowD. For sowD prior to DOY = 270, $NDVI_{Dec}$ remained at stable values; but decreased rapidly between DOY = 270 and 290; and remained at a relatively stable low value for sowD after DOY = 290. For 2017 (Fig. 6a) only a part of this relationship is apparent due to its smaller range of sowD. Using all field-level data points, the logistic relationship between sowD and $NDVI_{Dec}$ had an $R^2$ of 0.25 for 2017, 0.42 for 2018, and 0.55 for the combination of both years (Fig. 6c).

To evaluate the effect of temperature on $NDVI_{Dec}$, maize field groups with the same growth period were analysed and plotted in Fig. 6d. Within each group, warmer seasonal temperatures tend to result in higher $NDVI_{Dec}$ values, although this effect is most pronounced for sowD between DOY = 270 and 290. This indicates that a 10–20 day delay from optimal cover crop planting dates can be partially offset by higher autumn temperatures.

### 4.4. Optimal timing for sowing cover crop

The relationship between cumulative GDD and $NDVI_{Dec}$ was evaluated by combining data from both years (Fig. 7a). Using the fitted logistic curve, cumulative GDD could explain 50% of $NDVI_{Dec}$ variability of all analysed maize fields. The box-and-whisker plot groups data within intervals of 20 GDD units and demonstrates a positive relationship between $NDVI_{Dec}$ and cumulative GDD until reaching a

cumulative GDD of approximately 400. Prabhakara et al. (2015) also found that percent vegetation cover increased linearly until reaching 400 GDD (when using the same $T_{base}$ of 4 °C), but saturated after 400 GDD for six winter cover crops. Taking therefore 400 GDD as the required value to be reached by 1 December, we found that for the past three decades 18 September ( ± 7.2 days) and 19 September ( ± 7.1 days) are on average the optimal cover crop sowing dates according to temperature data recorded at weather station 'Twente' and 'Heino', respectively (Fig. 7b).

## 5. Discussion

The increasing availability of high-resolution satellite imagery at short repeat intervals offers potential for more accurate estimation of cover crop phenology and vegetative ground cover at field level, and consequently the link of the estimation to weather variables. This study takes advantage of the capability of Sentinel-2 to collect regular observational data for individual fields across a large area, showing that while winter cover crops have been established on nearly all maize fields, a large variability of winter ground cover between fields existed, along with a considerable range of cover crop sowing dates within the province of Overijssel in 2017 and 2018.

Our analysis confirmed that autumn temperature is an important explanatory factor for vegetative ground cover of cover crops. The winter ground cover of winter rye and Italian ryegrass in Overijssel was found to be sensitive to autumn temperatures, particularly when sowing took place between DOY 270 and 290 (Fig. 6d), i.e., the temporal range when ground cover decreases rapidly with sowing date (Fig. 6c). This suggests that increases in autumn temperature could benefit cover crop growth, possibly justifying later sowing dates. According to data from the weather stations in Overijssel, the mean October temperature has increased by 0.9 °C over the past 30 years (data not shown). In addition, autumn and annual mean temperatures are not uniform across the Netherlands (Ligtvoet et al., 2013). The KNMI'14 climate scenarios reported that the annual average warming in the southeast of the Netherlands is greater than in coastal areas (Tank et al., 2015). These regional temperature differences and changes, particularly for autumn, suggest that the same sowing date may not be equally applicable across the Netherlands, and that location-specific sowing strategies may be required to minimise nitrogen leaching.

Although we found that sowD could explain 55% of the variability in $NDVI_{Dec}$, additional farm management practices are important to attain a good winter ground cover (e.g., Treadwell et al., 2007). These practices include among others planting method, species choice, and seeding rate. Planting methods used in the study area include no-till drilling, conventional drilling and broadcasting. Relative to broadcast, no-till drilling and conventional drilling associated with better seed-soil contact have been shown to increase germination and cover crop establishment (MAWP, 2008). With regard to species choice, different effects on vegetative cover and weed control were found between grass (cereals) and legumes, as well as between grass/legume mixtures and monocultures (Lawson et al., 2015). Seeding rate is another factor that was reported to affect cover crop biomass production and weed suppression (Boyd et al., 2009; Brennan et al., 2009). Brennan and Boyd (2012) found that increasing seeding rate to three times the normal rate resulted in a 2.7-fold increase in cover crop density, and thus increased shoot biomass during December. In addition to the management practices directly related to the cover crops, fertilizer rates on the maize crop and maize-uptake of fertilizer may vary, thus affecting the remaining soil nitrogen that is available for the cover crop and as a consequence the cover crop productivity. Herbicide carryover from maize is another factor that leads to poor establishment and decreased autumn growth of cover crops, particularly for some species (e.g., crimson clover and Italian ryegrass; Cornelius and Bradley, 2017) that are more sensitive to residual herbicide than others. Despite these management factors, our study clearly demonstrates a dominant effect of cover crop sowing date on winter vegetative cover in the study area, which is consistent with other studies (Hively et al., 2009; Lawson et al., 2015).

Given the importance of the timing of cover crop sowing, from 2019 onwards the Dutch government implemented a new legislation that states that after maize cultivation on sand and loess soils a cover crop must be sown by 1 October. For Overijssel, we found for the past three decades 19 September ( ± 7 days) is on average the optimal sowing date based on the 400 GDD threshold (Fig. 7b), which provides technical support for the legislation. However, optimal dates varied considerably over the past 30 years, suggesting that in specific years a substantially earlier sowing date is required (e.g., 11 September in 2010) to attain 400 GDD by 1 December.

For farmers, there is a cost in sowing cover crops early. Because their main objective is to attain a good maize harvest, maize may not be fully ripe by 1 October, depending on weather conditions during the growing season. In addition, field accessibility and contractor availability may be limited. Therefore, we foresee that the practice of undersowing will gain in importance following the new regulation. With undersowing, the cover crop is sown simultaneously with or (more commonly) shortly after the establishment of the main crop (Peterson et al., 2019). The practice of undersowing a cover crop is already widespread in colder regions such as Nordic countries (Aronsson et al., 2016; Valkama et al., 2015). A number of studies have reported that undersowing cover crops into maize at leaf stages 4–7 does not result in large maize yield reductions, while providing abundant winter ground cover (Belfry and Van Eerd, 2016; Noland et al., 2018). However, the practice of undersowing also involves potential challenges for farmers. The primary difficulty with undersowing lies in the delicate balance of timing: the cover crop must be interseeded early enough to receive sufficient solar radiation prior to maize canopy closure, yet late enough to minimise the competition with maize for water, nutrients and light (Noland et al., 2018). We expect that the coming years will see a large increase of undersowing practices in Dutch maize systems for regulation compliance reasons, and experience will be gained to overcome the stated challenges. For our study period, undersowing was not applied in the great majority of fields. We do expect that Sentinel-2 time series can provide a distinct temporal pattern that allows undersowing and post-harvest sowing to be separated, mostly because of the higher greenness levels after the maize harvest (Fig. S3). As such, further analysis with satellite data could reveal the effectiveness of undersowing for winter ground cover in general, and possibly of different undersowing practices.

Our study demonstrated the effectiveness of using Sentinel-2 for acquiring observational data on sowing dates and winter vegetative cover for small agricultural fields. Nonetheless, we found that information retrieval was not equally effective across the study region. Limitations for deriving phenological estimates exist when fewer satellite observations in time are available, e.g., due to large temporal gaps in critical periods. Because Sentinel-2 has in principle a short (five day) revisit time, cloud cover is the main cause of such gaps. In this study, temporal gaps were more prominent for 2017 than for 2018, due to less cloud cover as well as the full benefit of the Sentinel-2A and -2B combination in 2018. We note however that a large part of the study area had overlapping orbits resulting in 2–3 day intervals and consequently a larger availability of cloud-free observations, which allows for a better temporal description of seasonal vegetation changes (e.g., Vrieling et al., 2018). For countries like the Netherlands this can be critical, given that there are 200 cloudy or rainy days on average per year and a large fraction of these occur during the growing season of cover crops from September to December (Osborn et al., 2016). In fact, most of the failed retrievals were outside the region with orbit overlap (Fig. 5). Nonetheless, for Overijssel retrievals were successful for 83% and 94% of the maize fields for 2017 and 2018, respectively. To remediate image paucity in specific regions, particularly for non-overlapping areas, Sentinel-2 image series could be combined with data

from other similar satellite sensors to increase temporal coverage. One option for this could the use of the harmonized Landsat and Sentinel-2 surface reflectance data set (Claverie et al., 2018). While we do not claim that our current Sentinel-2 based approach will work for all fields anywhere, it does allow to obtain observational evidence on temporal cropping information for relatively small fields ( > 0.5 ha) across wide geographical areas.

The current study presented a method for determining optimal cover crop sowing dates based on a cumulative GDD threshold obtained through assessing the effect of satellite-retrieved sowing dates on cover crop performance. We found that spatial differences of monthly mean temperature existed within one of the 12 Dutch provinces (Fig. S2), suggesting that the optimal cover crop sowing date differs geographically, particularly when focusing on a larger region (e.g., the whole of the Netherlands). Therefore, this study could be extended to the whole country through 1) collecting observational evidence for sowing dates and winter vegetative cover from Sentinel-2 analysis for a larger geographical area, and 2) a national analysis of when 400 GDD units are attained, for example to verify if the 1 October cut-off date is equally valid across the country. Such analyses may gain in relevance, given that a wider implementation of cover crop obligations for other non-maize production systems is envisaged under Dutch legislation. Applying the techniques for estimating the cumulative GDD threshold, further research may also be warranted to explore optimal sowing dates for alternative cover crop species (e.g., Japanese oats) and for other main crop systems that leave a relatively large amount of nitrogen in the soil after harvesting (e.g., potato, tulip and cruciferous vegetables such as cauliflower and broccoli), as well as better understanding the trade-offs between main crop harvest quality and winter cover crop coverage. Increasing the effectiveness of cover crops in reducing nitrogen leaching is complex, requiring management that is adapted to variable weather and soil conditions. This study provides one piece of this complex puzzle, by defining climatologically-sensible optimal sowing times. The real challenge remains with the farmers who have to balance a good economic profitability of the main maize crop, nutrient efficiency, and regulation compliance, while constantly adjusting and anticipating to climate variability.

## 6. Conclusions

An effective establishment of winter cover crop after maize cultivation is important for reducing nitrogen leaching to groundwater in the Netherlands. We showed that the effectiveness and timing of cover crop growth for small agricultural fields can be assessed from Sentinel-2 image time series across a wide landscape. While cover crops are grown on nearly all maize fields during autumn, we found a large variability of winter ground cover between fields within the province of Overijssel. This variability showed a strong negative logistic relationship with cover crop sowing date and a positive relationship with cumulative GDD until reaching approximately 400 GDD. Based on this threshold, we demonstrated that for the past three decades, 19 September ( ± 7 days) is on average the optimal sowing date in Overijssel. Given that maize ripening may often take place later than this date, maize-growing farmers in the Netherlands are expected to practice much more springtime undersowing, particularly because the new 2019 legislation requires a cover crop to be sown by 1 October. Further efforts are needed to evaluate how effective the current legislation is for control of nitrogen leaching across the Netherlands. Although these findings were obtained by taking advantage of the high-revisit frequency of Sentinel-2, our results reveal a lower success rate of phenology retrievals for areas not imaged by overlapping orbits. To obtain a denser temporal coverage in such areas, additional optical satellite sensors may need to be integrated. Nonetheless, this study demonstrated that Sentinel-2 series, given their fine spatial and temporal resolution, can result in actionable field-level information for farmers and legislators.

## CRediT authorship contribution statement

**Xinyan Fan:** Conceptualization, Methodology, Software, Formal analysis, Writing - original draft, Visualization. **Anton Vrieling:** Conceptualization, Methodology, Writing - review & editing, Supervision. **Bert Muller:** Investigation, Writing - review & editing. **Andy Nelson:** Writing - review & editing, Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jag.2020.102139.

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